AI + EDUCATION

From AI Tutors to A New Pedagogy for Learning in the AI-era

What should universities teach when expert, adaptive explanations are available to every student?

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Students are already using AI to study

57%

of surveyed U.S. college students use AI for coursework at least weekly.

Available anytime and (way too) powerful

e.g., subtlties, and multi-modal



Common phenomena: (1) attendence rate even lower (2) no one asks questions

4. A subtle error: counting sample patterns before symmetrization

A tempting shortcut is:

> "There are at most $\Pi_{\mathcal{H}}(n)$ prediction patterns on the sample. Pick one representative per pattern, then apply Hoeffding and a union bound."

This is not justified. The representatives depend on the sample, and hypotheses agreeing on that sample can have different population risks. Conditioning on the entire sample does not fix this: it leaves no sample randomness to which Hoeffding could apply.

The correct argument separates three steps:

1. **Symmetrize in expectation.** An independent ghost sample and exchangeability give

$$\mathbb{E}_S \sup_h \big(L(h) - \widehat{L}_S(h)\big) \le 2\,\mathbb{E}_S \widehat{\mathfrak{R}}_S(\mathcal{F}),$$

   where $\mathcal{F}$ is the loss class. This is an expectation inequality, **not a sample-by-sample bound**.

2. **Condition, count, then average.** Conditional on S, the Rademacher signs remain independent and random, and identical loss vectors give identical signed sums. Massart's and Sauer's lemmas therefore give

$$\widehat{\mathfrak{R}}_S(\mathcal{F}) \le b_n := \sqrt{\frac{2d\log(en/d)}{n}}.$$

   Since this holds for every S, averaging gives $\mathbb{E}_S \widehat{\mathfrak{R}}_S(\mathcal{F}) \le b_n$, with no additional factor.

3. **Concentrate over samples.** Bounded-differences concentration turns the expected-gap bound into a high-probability bound, adding a confidence term.

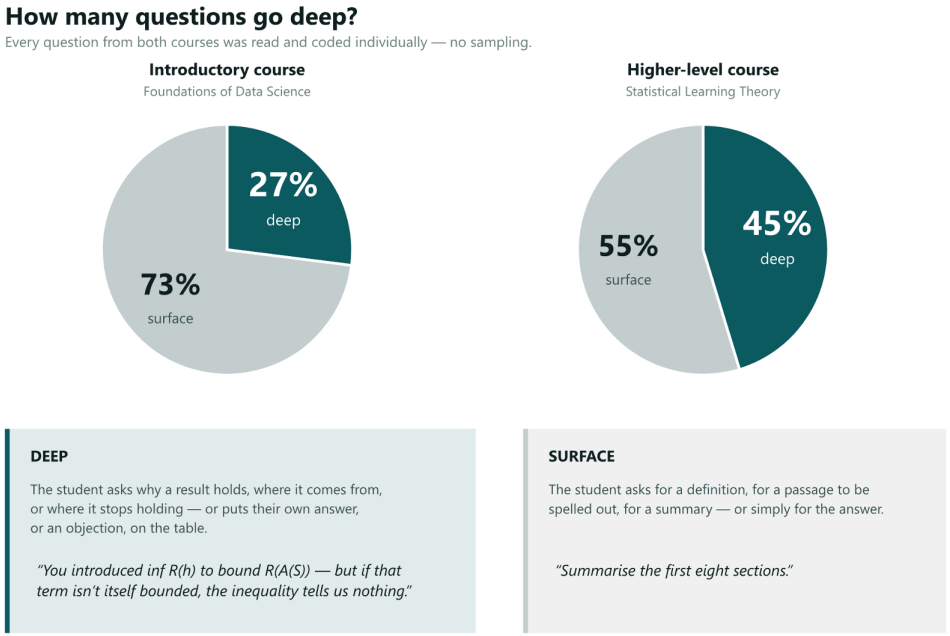
The factor 2 comes from symmetrization; removing the conditioning costs nothing.

Lumina–Gallup 2026: 57% of surveyed U.S. college students use AI for coursework at least weekly; Bastani et al., PNAS 2025

A powerful AI does not necessarily help learning

Multiple studies showed that learning **with** access to AI may lead to worse independent test performance (i.e., **without AI**).

- Bastani *et al.*, "Generative AI without guardrails can harm learning: Evidence from high school mathematics." PNAS, 2025
- Kosmyna *et al.*, "Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task." arXiv preprint 2506.08872
- Stadler *et al.*, "Cognitive ease at a cost: LLMs reduce mental effort but compromise depth in student scientific inquiry." Computers in Human Behavior, 2024.



Question depth varies across courses, and surface fluency can hide shallow learning.

Bastani et al. (2025); Kosmyna et al. (2025); Stadler et al. (2024); Tao, Mathematics in the Age of AI (2026)

Why assisted learning can miss the learning goal

Most observed questions are surface questions

Translation, explanation, summarize/quick-learn, and answer-course-test-question requests dominate the observed interactions.

Possible explanations

1 · Incorrect use of AI

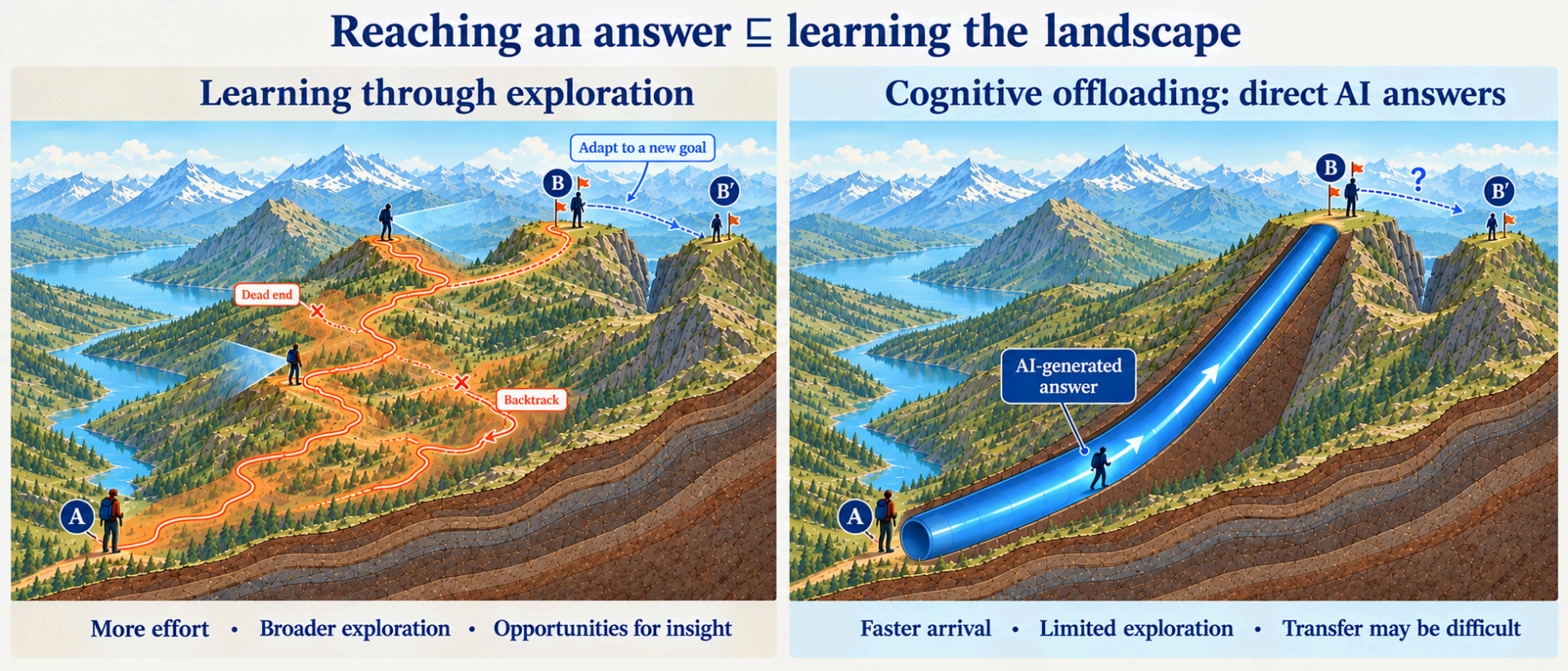
Lack of **active retrieval** and **productive struggle**.

2 · Partial assessments

Current assessments cannot fully assess capabilities developed by AI-assisted learning, such as system thinking and dealing with complexities.

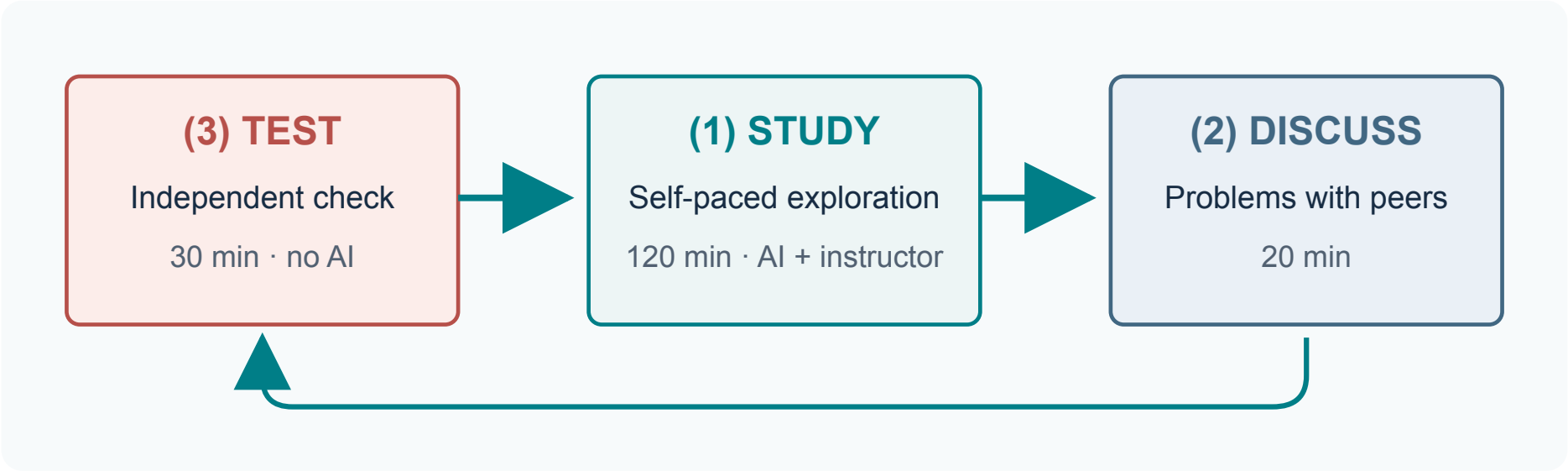
TDAA question-depth coding; curriculum interpretation

Exploration teaches what an answer cannot



Similar to [Tao's argument between Math 1.0 and 2.0]

TDAA: Learning and exploring in class



- **(1) Self-paced learning**: cater for different types/learning-styles of students
- **(1+2) AI, instructors, and classmates**: offer a learning community, feedback, alternative views, emotional support, and guided exploration, where solitary study **rarely** provides.
- **(3) Internalization**: coerced by the test

The test anchors mastery without limiting exploration

1. **Reduce procrastination.** Key for courses that have deep dependency.
2. **Reward internalization.** Sense of achievement.
3. **Test core understanding.** Questions target the average expected level; exploration is not examined.
4. **Leave room to explore.** 120/130 points on paper, but capped at 100.

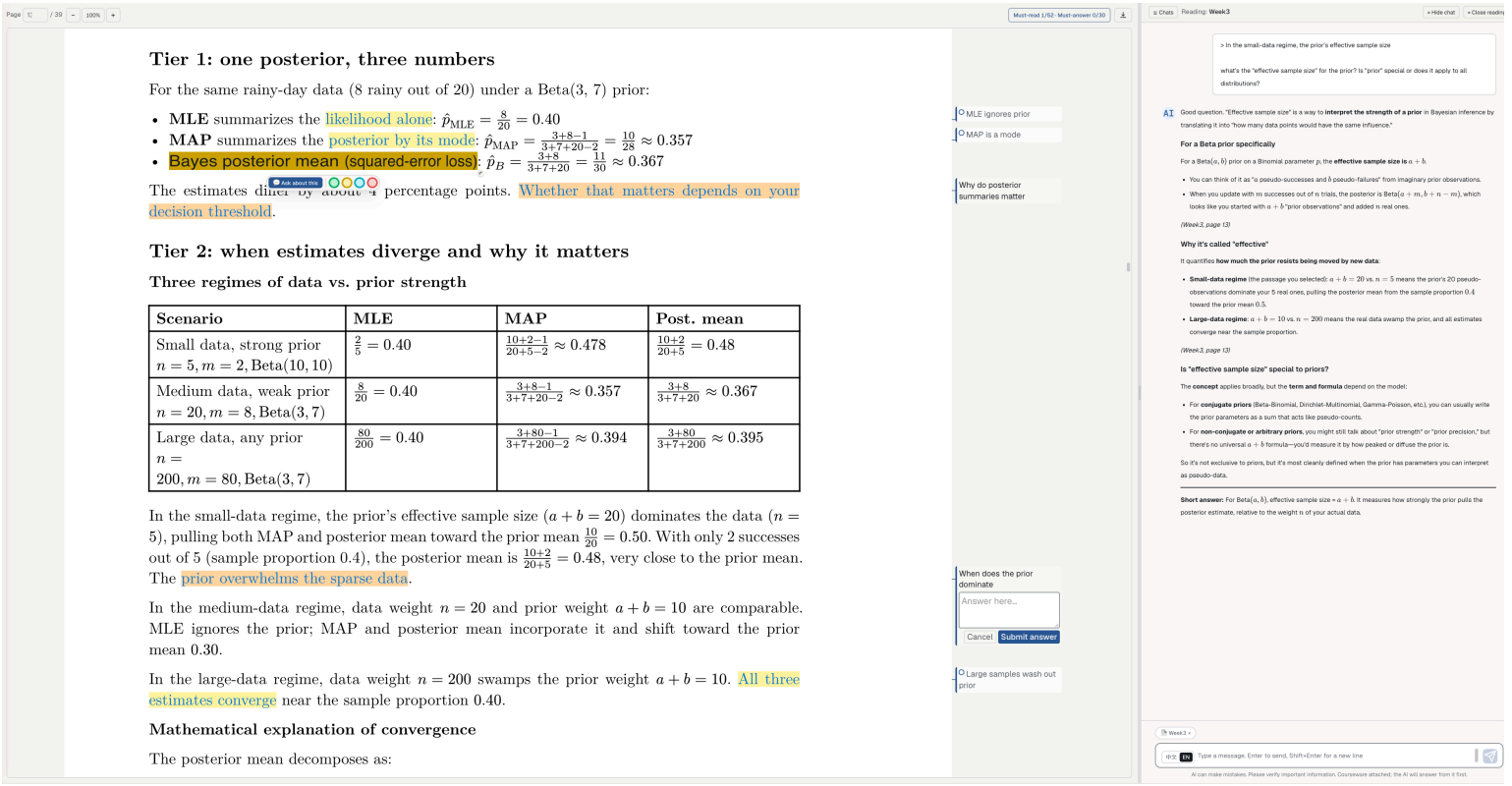
AI-generated materials and AI platform

Materials

- Prose guided by questions, examples, and applications
- A pedagogy playground: three tiers, scaffolding, and alternative materials

AI platform

- Personal highlighting and notes
- Instructor “to read / to answer” tasks
- Contextual questions and answers



Tier 1: one posterior, three numbers

For the same rainy-day data (8 rainy out of 20) under a Beta(3, 7) prior:

- **MLE** summarizes the likelihood alone: $\hat{p}_{\text{MLE}} = \frac{8}{20} = 0.40$
- **MAP** summarizes the posterior by its mode: $\hat{p}_{\text{MAP}} = \frac{3+8-1}{3+7+20-2} = \frac{10}{28} \approx 0.357$
- Bayes posterior mean (squared-error loss): $\hat{p}_B = \frac{3+8}{3+7+20} = \frac{11}{30} \approx 0.367$

The estimates differ by about 4 percentage points. Whether that matters depends on your decision threshold.

Tier 2: when estimates diverge and why it matters

Three regimes of data vs. prior strength

Scenario	MLE	MAP	Post. mean
Small data, strong prior $n = 5, m = 2, \text{Beta}(10, 10)$	$\frac{2}{5} = 0.40$	$\frac{10+2-1}{20+5-2} \approx 0.478$	$\frac{10+2}{20+5} = 0.48$
Medium data, weak prior $n = 20, m = 8, \text{Beta}(3, 7)$	$\frac{8}{20} = 0.40$	$\frac{3+8-1}{3+7+20-2} \approx 0.357$	$\frac{3+8}{3+7+20} \approx 0.367$
Large data, any prior $n = 200, m = 80, \text{Beta}(3, 7)$	$\frac{80}{200} = 0.40$	$\frac{3+80-1}{3+7+200-2} \approx 0.394$	$\frac{3+80}{3+7+200} \approx 0.395$

In the small-data regime, the prior's effective sample size ($a + b = 20$) dominates the data ($n = 5$), pulling both MAP and posterior mean toward the prior mean $\frac{10}{20} = 0.50$. With only 2 successes out of 5 (sample proportion 0.4), the posterior mean is $\frac{10+2}{20+5} = 0.48$, very close to the prior mean. The prior overwhelms the sparse data.

In the medium-data regime, data weight $n = 20$ and prior weight $a + b = 10$ are comparable. MLE ignores the prior; MAP and posterior mean incorporate it and shift toward the prior mean 0.30.

In the large-data regime, data weight $n = 200$ swamps the prior weight $a + b = 10$. All three estimates converge near the sample proportion 0.40.

Mathematical explanation of convergence

The posterior mean decomposes as:

MLE ignores prior

MAP is a mode

Why do posterior summaries matter

When does the prior dominate

Answer here...

Cancel Submit answer

Large samples wash out prior

Reading: Week 3

Hide chat

> In the small-data regime, the prior's effective sample size

what's the "effective sample size" for the prior? Is "prior" special or does it apply to all distributions?

AI Good question. "Effective sample size" is a way to **interpret the strength of a prior** in Bayesian inference by translating it into "how many data points would have the same influence."

For a Beta prior specifically

For a Beta(a, b) prior on a Binomial parameter p, the **effective sample size is** $a + b$.

- You can think of it as "a pseudo-successes and b pseudo-failures" from imaginary prior observations.
- When you update with n successes out of n trials, the posterior is Beta$(a + m, b + n - m)$, which looks like you started with $a + b$ "prior observations" and added n real ones.

(Week3, page 13)

Why it's called "effective"

It quantifies **how much the prior resists being moved by new data:**

- **Small-data regime** (the passage you selected): $a + b = 20$ vs. $n = 5$ means the prior's 20 pseudo-observations dominate your 5 real ones, pulling the posterior mean from the sample proportion 0.4 toward the prior mean 0.5.
- **Large-data regime**: $a + b = 10$ vs. $n = 200$ means the real data swamp the prior, and all estimates converge near the sample proportion.

(Week3, page 13)

Is "effective sample size" special to priors?

The **concept** applies broadly, but the **term and formula** depend on the model:

- For **conjugate priors** (Beta-Binomial, Dirichlet-Multinomial, Gamma-Poisson, etc.), you can usually write the prior parameters as a sum that acts like pseudo-counts.
- For **non-conjugate or arbitrary priors**, you might still talk about "prior strength" or "prior precision," but there's no universal $a + b$ formula—you'd measure it by how peaked or diffuse the prior is.

So it's not exclusive to priors, but it's most cleanly defined when the prior has parameters you can interpret as pseudo-data.

Short answer: For Beta(a, b), effective sample size $= a + b$. It measures how strongly the prior pulls the posterior estimate, relative to the weight n of your actual data.

Week 3

Type a message. Enter to send, Shift+Enter for a new line

AI can make mistakes. Please verify important information. Courseware attached: the AI will answer from it first.

Course platform screenshot: input/AI-platform-1.png; anonymized example

The importance of the instructor

Offline

1. **Architect the learning**
2. **Generate learning materials**
3. **Prepare and collect good questions**

Online

1. **Support the learner mentally.** Notice frustration, restore confidence, and sustain motivation.
2. **Guide learning and exploration.** Check understanding, give suggestions, answer and solicit questions.
3. **Read the room.** Decide when to intervene with a mini-lecture, discussion, or student sharing.



The instructor is the composer and conductor; let the class be water.

The importance of the instructor

Why will a student want to learn something *hard*?

1. **Love the teacher** (their passion for knowledge)
2. **Has an interest** in the course content
3. **Has a need** that the course can (partially) fulfill
4. **Has a plan/dream**



The instructor is the key — they can influence all four fundamental motivations.

On exploration: teaching takes more preparation

AI or AI-assisted students can raise questions **beyond the textbook**. Instructors need **full-stack knowledge, preparation, and on-the-fly learning** with AI.

Diverse perspective/routes

Proof of RREF's uniqueness:

1. Construct the canonical form from the row space (textbook).
2. Assume two different reduced forms; derive a contradiction (student).
3. Use a dual-basis perspective (AI under prompting).

Great questions

- Why does the Halting Problem proof not apply to the halting problem for DFAs?
- A probability space has disjoint outcomes. How should we model people's menu choices?
- When extending a finite-class PAC bound to an infinite class, can the growth function $\Pi(\mathcal{F})$ replace $\mathcal{H}$, rather than using Rademacher complexities?



An underprepared or passive instructor may reasonably appear less helpful than AI.

Illustrative proof paths for uniqueness of reduced row echelon form; questions from learning and exploration

Version 1 feedback points to a broader Version 2

TDAA VERSION 1 · SURVEY RESULTS DSAA3073 · Theory of Computation

10 of 18 respondents reported learning both deeper and faster.

Six reported deeper but slower learning; two reported similar or worse. These are student perceptions.

TDAA VERSION 2 · TESTING DIFFERENT IDEAS

Design changes

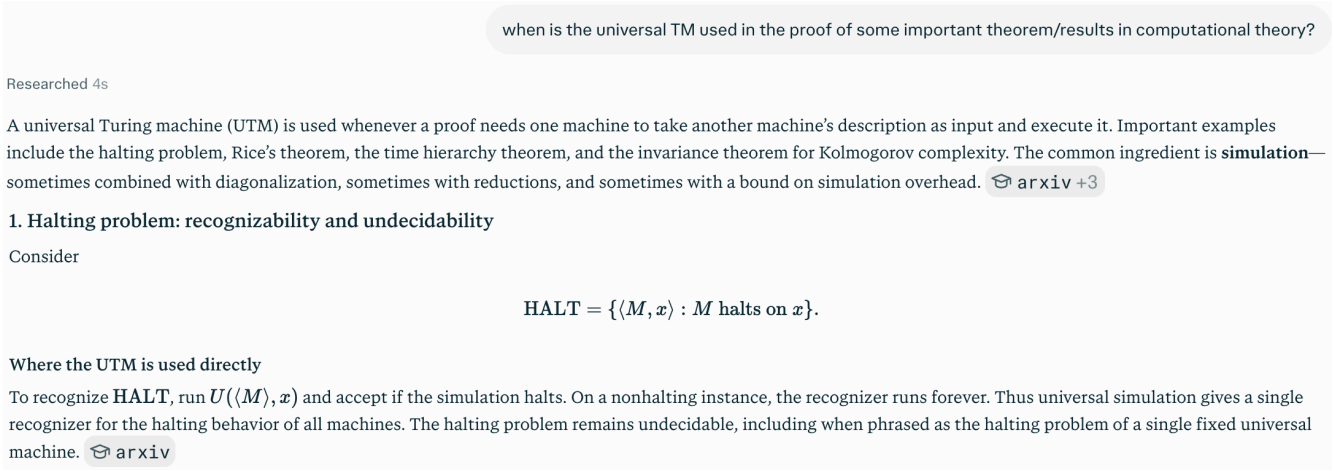
- Adapt to different cohorts of students & course content
- Fewer tests; more time for depth
- More in-class learning activities
- Ask students to generate: questions, discussion, mind maps
- Test alternative pedagogies for generated materials
- Trace finer-grained learning activities

Courses represented

- DSAA1001: Introduction of Data Science and Analytics
- DSAA3073: Theories in Data Science
- UFUG1601: Introduction to Computer Science
- UFUG2103: Linear Algebra

DSAA3073: Theory of Computation survey; Version 2 courses listed on slide

Hone higher-order skills



when is the universal TM used in the proof of some important theorem/results in computational theory?

Researched 4s

A universal Turing machine (UTM) is used whenever a proof needs one machine to take another machine's description as input and execute it. Important examples include the halting problem, Rice's theorem, the time hierarchy theorem, and the invariance theorem for Kolmogorov complexity. The common ingredient is **simulation**—sometimes combined with diagonalization, sometimes with reductions, and sometimes with a bound on simulation overhead. arxiv +3

1. Halting problem: recognizability and undecidability

Consider

$$\mathrm{HALT} = \{\langle M, x\rangle : M \text{ halts on } x\}.$$

Where the UTM is used directly

To recognize $\mathbf{HALT}$, run $U(\langle M\rangle, x)$ and accept if the simulation halts. On a nonhalting instance, the recognizer runs forever. Thus universal simulation gives a single recognizer for the halting behavior of all machines. The halting problem remains undecidable, including when phrased as the halting problem of a single fixed universal machine. arxiv

Take **critical thinking** as an example:

1. AI-generated sufficient number of effective training examples when learning the knowledge
2. Transfer to human or AI generated contents

💡 Knowledge is cheap; hone the skills$\Longrightarrow$ select content to teach

DFS-style Learning

Bound the probability of some rare event (typically $|X - \mathbb{E}[X]|$)

1 · CLT
Asymptotic bound

2 · Tail inequalities
Markov, Chebyshev, Hoeffding

3 · Extend the proof technique
(Upper) bound $p(x) \Longrightarrow$ optimization problem

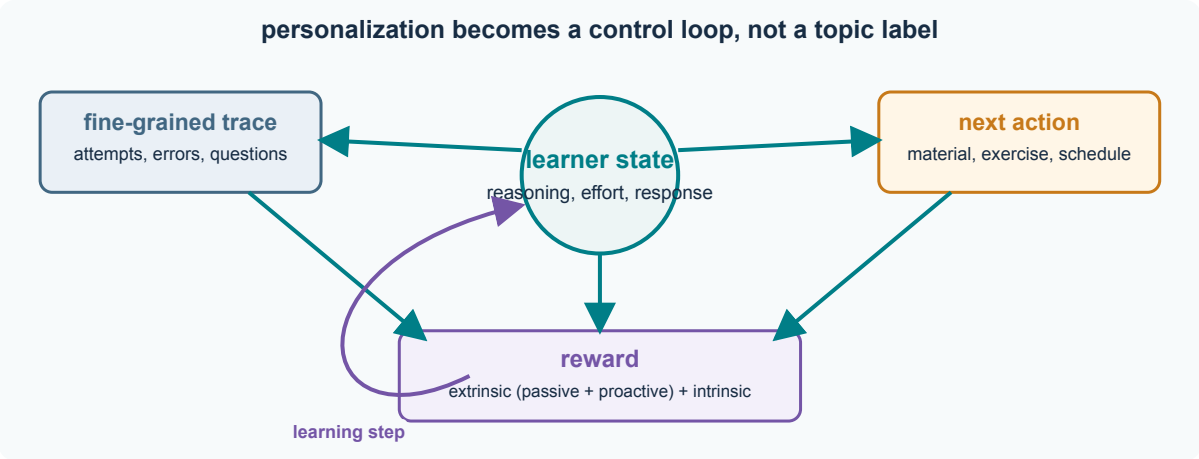
4 · Experiment
(Numerically) find the worst-case for a tight bound

Deeper: D. Bertsimas, I. Popescu. "*Optimal Inequalities in Probability Theory: A Convex Optimization Approach*.", SIAM Journal on Optimization, 2005.



This is depth-first learning, or **mini-research** (enabled by AI): solve a real question, learn the needed tools, explore in depth and make associations; periodically patch and consolidate.

Extremely personalized learning



Reinforcement-learning-style learning

- **State:** reasoning, effort, and response.
- **Trace:** attempts, wrong steps, rereading, questions, and feedback responses.
- **Next action:** material, exercise, depth, and timing based on current evidence.
- **Reward:** based on the trace, next action, and state; combine progress, AI feedback, and curiosity.

💡 The current learning system is inefficient; choose a locally optimal policy to improve learning efficiency significantly.

Personalized learning as a reinforcement-learning-style control loop

Conclusions

AI changes both how students learn and what they need to learn.

Scientific research into AI + education

- Exploit AI's capabilities, while keeping the learner's reasoning visible.
- Understand and model how people learn, struggle, and transfer.
- Use technical and scientific methods: build, compare, measure, falsify.
- Develop and iterate quickly as the models change.

💡 Call for motivated instructors to join us to try out new ideas in real courses.